

1st Examination Date Summer Semester 2026

Lecturer: Benjamin Groß and Prof. Dr. Stephan Huber

Title of examination: Analytical Skills for Business

Study programme: MBA90e, MIMe

Semester: 2

Date: 27.04.2026

The following sections need to be completed legibly by the student:

Name/

Matriculation number:

Alireza Teytounchi 400889526

Study programme:

International Management (M.A.)

DECLARATION OF THE STUDENT ON HIS/HER ABILITY TO TAKE THE EXAMINATION:

The student declares that there are no restrictions regarding his/her mental, emotional and physical state and that he/she is fully capable of taking the examination in its full extent.

Grade:

NOTA BENE: PRELIMINARY REMARKS

- Time limit: **90 Minutes**
- Examination aids: **Calculator**
- Mobile phones, Smartwatches and any kind of web-enabled and programmable devices need to be switched off before the beginning of the examination and are not allowed near body and desk!
- The task sheets need to be handed in with your answers.
- If there is more than one lecturer responsible for the examination, questions need to be answered on separate exam sheets.
- All exam sheets that have not been used need to be handed back to the exam supervisor.
- A date for a review of the marked examination paper will be communicated via ILIAS.

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Studiengang / Fachsemester Study program / Subject semester

Analytical Skills for Business.

Prüfungsfach / Prüfungsdozent:in Examination subject / Examiner

1): a): The branches is a copy from main project that we could create before starting the project for working parallel at the same time and different individual can working on it at the same time without any errors.

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- and with the pull request you could update your information and codes on your branch and for any pull requests an individual share the last updates.

- This workflow ^{concret example} support collaboration, because it's a cloud-based platform that give access to the different persons who are working on one project and this is an advantage of branches that permit collaborate your data.

b): version Control System (VCS) that helps track and changes for files over the time. and if we don't use the version control system it can lead to misunderstanding or choose.

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Risk on the Collaboration and Branching or merging

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2) a) categorical data, is define the data that are showing the different type of datas. ~~data~~ included some string and numerical. like

using scatter plot for

Student	Age	Score
Student1	29	100
Student2	31	97

Visualization and
bar plot for occurrences

Comparing the categories. (Comparison between 2 value)

Nomirical data, is define the data that is just Numbers. integer and Float. like: 2, 4, 6, 8.

histogram for instance 2.1, 3.8, 4.12.

Using bar Plot for visualization and Comparin son between the value of numbers.

Time-series, is define the data that based on one category at the one point time. like: the Sales in 2024 From ~~different~~ different companies. But in one time.

Company	1	2	3
Sales 2024	400	450	500

using Pie chart for visualization and Comparison between the value at the one point time.
not used in serious data science

The visualization give the visual comparison to the stakeholders to know and gain the information for comparison and this method is very usefull and easy to make better decision.

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b) Scatter plot: whats the differences between variables? and based on the scatter plot the managers could seek the differences.

② Histogram: How much does we sell in april and what's

the sales in last four months?

Based on histogram plot we represent the values of variables on one page with one plot. and we could show the range of the sales on the plot for managers. *examples?*

③ Bar-chart:

3): a): Data cleaning workflow includes different steps such as:

① Profiling: according to the profiling the manager or data scientist sort the data based on categories. *not only?*

② Validation: in this step the scientist examine the data that is this data has a positive affect and increase the rate of process. they examine the variables based on values.

schema, ranges, rules, business etc.

③ Cleaning: in this step the scientists have to normalize the data and clean the incomplete, inconsistent, outliers, Duplicate the data that lead to better analysis and make better decision.

④ Verification: in this step we have to apply the data, documents, information that is useful for continue the process. This step is very important, because this is the last step for ensuring that the data are ready to documentation.

is the outcome as expected?

⑤ Documentation: This is the last part of data cleaning workflow, that ~~starting~~ starting the analyzing with data that is normalized and here dividing the data in categories to start analyzing.

documenting the data cleaning process

in correct information → Delete, remove duplicate, remove or delete outliers, Delete data that consist errors.

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3) b): Duplicates:

→ cleaning step

for example some data that repeat in data set with the same name, value, variable most to delete them.

we ~~most~~ most to delete or remove or combine the Duplicates to one data.

The strategy is Normalizing data to delete that value are repeated.

only 1 example?

5/10.

10/20

4) b): SVM. Based on linear regression we could Predict the data that is linear and find the solution for the future decision-making.

without some prediction, we couldn't insights and vision and the purpose is unclear.

So some predictive tools are important for the decision making and choose the best solution for the company.

concrete example missing

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a): Correlation coefficient's measures define the value of the utility and conditions - to improve it we could using the -

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5) a) n8n is a cloud-based platform automation tools that is something low-code or no-code. This platform is online and based on this application the individual can connect the different app together with trigger and just drag and drop and some low coding. **5/5**

⊗ Create work flow → Trigger → Actions
For example: Receive an email → Copy on the Notepad → update the email or send it on the Telegram.

5) b) no-code: create a program without any knowledge of programming. just with visualization and drag and drop tools and connect them. Like: Air Table - Bubble

Low code: create a program with a little coding + drag and drop.

in the no code an individual ~~can~~ without any knowledge can create a program. and in low-code the programmer is more flexible because in this type the individual can write and create a powerful app. **8/10**

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6) Library (Tidyverse) **tibble!**

```
xpi_data <- c("Jan", "Feb", "Mar", "Apr")
```

```
xpi_data <- c(42, 50, 47, 55)
```

or

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```
kpi_data <- mutate(month, revenue)
```

```
kpi_data kpi_data <- mean(kpi_data$Revenue)
```

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```
b) library(ggplot2)
```

```
kpi_data <- (ggplot2)
```

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